Spring 2013 / EC571

Instructor:Ajay Joshi

Exam 1

Date: February 27, 2013

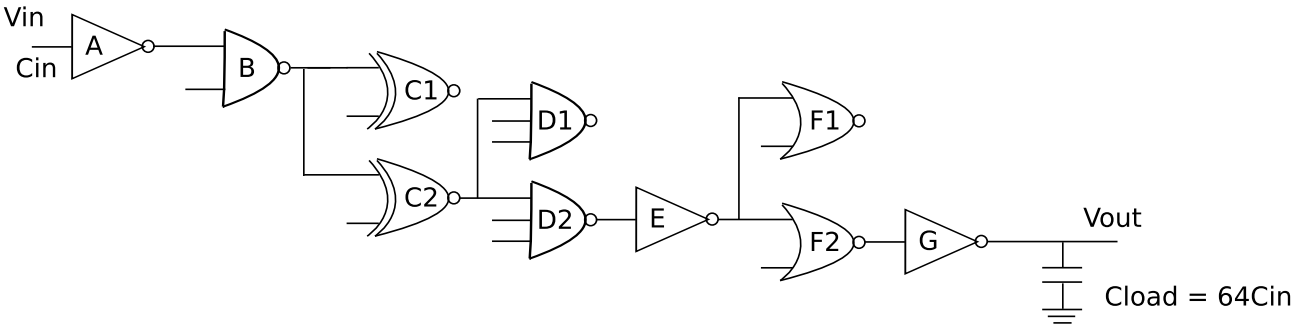
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours
2. The maximum grade is 100 points
3. Please write your name on all pages
4. The solutions should be written in the spaces provided on the test sheet
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number
6. This is an open book and open notes exam.
7. Material sharing is not allowed
8. Calculators are allowed
9. Boston University Academic Code of Conduct applies

Question 1 (20 points)



Using the logical effort technique, determine the input capacitances for the various gates in the data path from Vin to Vout for the logic design shown above. You can assume that C1 and C2 have the same sizes, D1 and D2 have same sizes and F1 and F2 have the same sizes. (20 points)

Question 2

1. Draw the static complementary CMOS logic implementation for the following logic - $f(A,B,C,D,E,F) = \overline{A \cdot (B+C) \cdot (D+E+F)}$. (5 points)

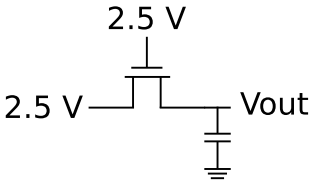
2. Design the equivalent RC network for the circuit. You can assume the circuit drives a load capacitance of C_L (5 points)

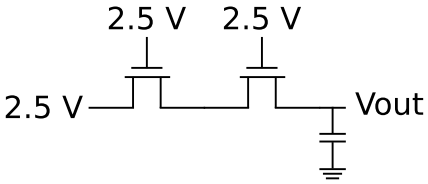
3. Determine the widths of the various transistors as a function of W so that worst case $\tau_{HL} = \tau_{LH}$. You can assume $2R_{sq}(NMOS) = R_{sq}(PMOS)$. (5 points)

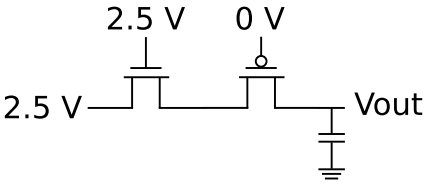
4. Calculate τ_{HL} and τ_{LH} using the following parameters – W = 1.5 micron, L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron, C_L = 50 fF. (5 points)

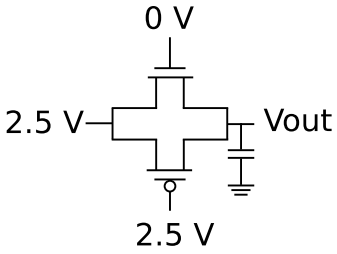
Question 3

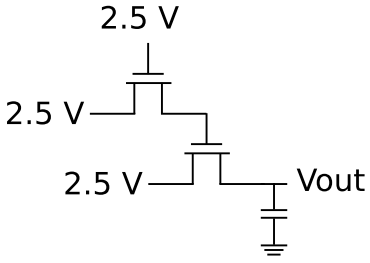
Find the final value of the voltage V_{out} for the following 5 configurations (4 points each). Assume V_{dd} = 2.5 V, $V_{th,N}$ = 0.5 V and $V_{th,P}$ = -0.5 V. Assume that the capacitor is initially discharged. Ignore subthreshold conduction and body effect.





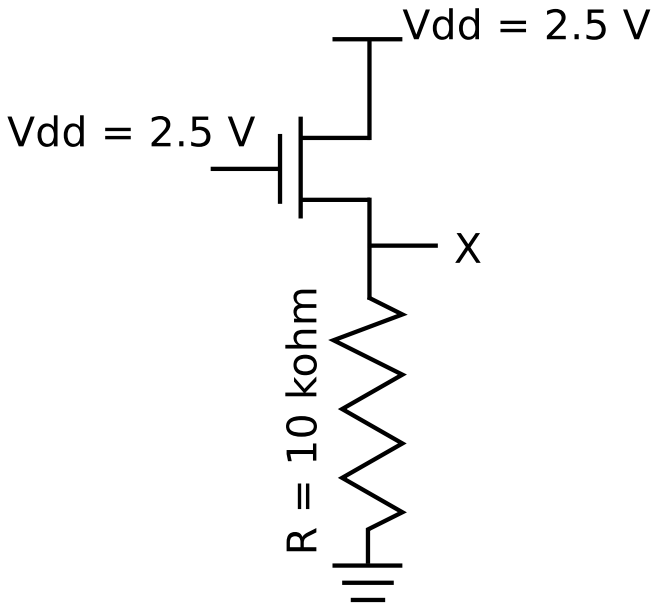






Question 4

Consider the circuit configuration given below. The NMOS has the following parameters : $V_{T,N}$ = 0.5 V, $\mu \cdot C_{ox}$ = 60 x 10^{-6} A/V^2. We would like to achieve a voltage of 1.0 V at node X in the circuit.



1. For V_X = 1.0 V, what would be the region of operation for the NMOS transistor? Explain how you determined the region of operation. (4 points)

2. Write down the equations that can be used to determine the current through the NMOS transistor. Ignore channel length modulation. (3 points)

3. Determine the required width of the transistor (for L = 0.25μm) to achieve V_X = 1.0 V. (6 points)

4. If the NMOS transistor was velocity saturated then determine the required width of the transistor (for L = 0.25μm) to achieve V_X = 1.0 V. Ignore channel length modulation. Use unified transistor model. V_{DSAT} = 1.0 V (7 points)

Question 5

Design the static complementary CMOS implementation of a 1-bit half adder circuit operating at 2 GHz with both sum and carry outputs driving a load of 100 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. *Hint: A 1-bit half adder has two 1-bit inputs – A and B and it generates two 1-bit outputs – SUM and CARRY.* (20 points)

Bonus question (5 points)
Prof. Joshi has to travel to international workshops and conferences to present his research. To be able to communicate with the locals, Prof. Joshi would like to learn how to greet in different languages. Please indicate the equivalent of 'Hello' in different languages. For example, one answer would be Hola (Spanish) (1 point per language.).